Project Test Plan



ACE

CYBER R&D Program

Conversational AI Interface for
Analyst-Driven Threat Investigation

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2025

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Document ID:	PTP-001
Version #:	1.3
Status:	Draft
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Date:	2025-11-20

1.0 Objective

This document outlines the test plan for evaluating the performance of the Conversational AI Interface (CAI) integrated with Elasticsearch within a controlled AWS-based Elastic Cloud Enterprise (ECE) testbed.

The primary objective is to execute controlled experiments to evaluate whether the CAI generates accurate Elasticsearch JSON DSL queries for threat-hunting tasks and whether those queries correctly identify pre-tagged threat events in a fully synthetic dataset.

1.1 Hypothesis (from PC-001)

If a Conversational AI interface is developed and integrated with Elasticsearch to support natural-language threat-hunting queries, then, when evaluated against a pre-tagged dataset containing failed-login and lateral-movement events, it will achieve an overall F1 score of at least 0.95 across all evaluated threat-hunting tasks, indicating balanced precision and recall in detecting threat-tagged log events.

1.2 Independent Variables (from RQH-001)

- Threat-hunting task type (failed logins, suspicious IPs, lateral movement)
- Natural-language query phrasing variations
- Model parameters used by Ollama (prompt template, task mode)

1.3 Dependent Variables (from RQH-001)

- F1 Score
- Precision / Recall
- Detection accuracy (percentage of retrieved pre-tagged threats)
- JSON DSL query validity rate

2.0 Scope

2.1 In-Scope

Testing of:

- Elasticsearch (ECE-hosted, version 9.1.5)
- Conversational AI Interface hosted on Ubuntu 22.04 Streamlit frontend
- Ollama + Mistral 7B Q4_K_M model for NL→DSL translation

Execution of:

- Failed login detection
- Suspicious IP correlation (GeoLite2-based synthetic IP pools)
- Lateral movement detection

Measurement of:

- F1 score, precision/recall against PI-Key ground truth
- Query validity rates
- Detection accuracy
- Query latency

Testing Environment:

- All tests will be run on the specific environment defined in the Testbed Configuration Plan (TBSCP-001).

2.2 Out-of-Scope

The following is out of scope relating to this research:

- Hardware benchmarking
- Real-world SOC traffic analysis
- Penetration testing or vulnerability exploitation
- Long-term load/stress testing
- Real PII/CUI or operational telemetry (all synthetic)
- Automated remediation or alerting workflows
- Comparison to human SOC analysts (as noted in testbed plan)

3.0 Test Environment

All testing will be conducted in accordance with the established Testbed and its defined configuration as noted in TSCP-001.

3.1 Key Components

Hostname	Role	IP Address	Key SW / Config
ece-primary	ECE Primary	10.120.16.100	RHEL 8.10, Podman containers (ES/Kibana 9.1.5)
ece-allocator	ECE Allocator	10.120.16.200	RHEL 8.10, Podman
log-gen-workstation	Log Generator & Filebeat Node	10.120.16.10	RHEL 8.10, Python 3.11, Faker, Filebeat
cai-node	Conversational AI Interface	10.120.29.230	Ubuntu 22.04, Streamlit, Ollama, Mistral 7B
analyst-workstation	Analyst UI Access	External/Public	Web browser, Kibana + CAI access

4.0 Prerequisites / Entry Criteria

Before any test case is executed:

1. All EC2 instances must be running within the testbed VPC (10.120.16.0/20).
2. Elasticsearch, Kibana, Filebeat services should be confirmed active.
3. All PI-Key reference files should be generated (generated-soc-logs-key.txt).
4. Baseline resource usage stable for at least 5 minutes:
    - CPU < 5%
    - Memory < 10%
5. The Conversational AI Interface must be running.
6. Researcher must have the Lab Notebook (LN-001) open for real-time documentation.

5.0 Test Cases

Test Case ID: *TC-001*

Title: Baseline – CAI Idle / No Query Issued

Objective: Establish baseline Elasticsearch resource usage with no queries.

Step-by-Step Procedure:

1. Verify entry criteria.
2. On ece-primary run:
   "dstat -tcmnd --output /data/raw/TC-001_dstat.csv &"
3. No traffic or CAI queries issued.
4. Observe for 5 minutes.
5. Stop data collection.
6. Verify output files exist.

Data to be Collected:

- dstat CSV
- Elasticsearch logs
- System metrics

Expected Outcomes / Success Criteria:

- CPU remains < 10%.
- No anomalies in Elasticsearch logs.
- All files generated successfully.

Test Case ID: *TC-002*

Title: CAI Natural-Language Query – Failed Logins

Objective: Evaluate CAI's ability to convert natural-language queries for "failed logins" into correct JSON DSL.

Step-by-Step Procedure:

1. Verify entry criteria.
2. Analyst enters the query:
   "Show me all failed login attempts in the past 24 hours."
3. Capture CAI output JSON.
4. Send JSON to Elasticsearch.
5. Retrieve event UUIDs and compare with PI-Key (ground truth).
6. Compute precision, recall, F1.

Data to be Collected:

- CAI-generated JSON
- Elasticsearch response UUID set

- PI-Key matched UUIDs
- Timing & error logs

Expected Outcomes / Success Criteria:

- Query is syntactically valid.
- F1 ≥ 0.95 for failed-login threats. (Hypothesis benchmark)
- No prohibited or unsafe query expansion.

Test Case ID: *TC-003*

Title: Suspicious IP Lookup

Objective: Test CAI's ability to generate correct DSL for correlating events involving suspicious IPs.

Step-by-Step Procedure:

1. Analyst enters:
   "Find events from foreign IP addresses within the past 2 hours."
2. CAI outputs JSON using field source.ip and adversary pool filters.
3. Compare output to PI-Key entries tagged: (threat).

Data to be Collected:

- CAI-generated JSON
- Elasticsearch response UUID set
- PI-Key matched UUIDs
- Timing & error logs

Expected Outcomes / Success Criteria:

- F1 ≥ 0.95
- CAI does not broaden query to unrelated fields.

Test Case ID: *TC-004*

Title: Lateral Movement Chain Detection

Objective: Validate CAI's ability to detect multi-step RDP/SSH lateral movements.

Step-by-Step Procedure:

1. Analyst enters:
   "Show me foreign lateral movement between hosts within the past hour."
2. CAI generates DSL matching multi-step adversarial chains.
3. Compare with PI-Key labels for lateral_movement_*.

Data to be Collected:

- CAI-generated JSON
- Elasticsearch response UUID set
- PI-Key matched UUIDs
- Timing & error logs

Expected Outcomes / Success Criteria:

- F1 ≥ 0.90 (slightly lower allowed due to multi-field complexity)
- Correct host-to-host correlation
- Query structure preserved (must use AND/sequence logic)

Test Case ID: *TC-005*

Title: Invalid / Ambiguous Query Handling

Objective: Ensure CAI safely handles malformed analyst inputs.

Step-by-Step Procedure:

1. Analyst enters ambiguous prompt:
   "Find the suspicious things but not too many."
2. CAI must either:
   - Request clarification, or
   - Fail safely with no DSL output.

Data to be Collected:

- CAI generated clarification response

Expected Outcomes / Success Criteria:

- No unsafe query generated
- No full-index wildcard scan (match_all)
- Safety guardrails triggered

Test Case ID: *TC-006*

Title: Query Robustness – Synonym & Paraphrase Variation Test

Objective: Evaluates the linguistic robustness of the CAI by giving minor variations in natural-language phrasing to determine if it degrades the detection accuracy.

Step-by-Step Procedure:

1. Verify Entry Criteria
2. Issue the following semantically equivalent queries into the CAI interface:
    - "Show me all unsuccessful login attempts from the past hour."
    - "List authentication failures for user accounts within the past hour."
    - "Which events had login problems in the last hour?"
    - "Display events in the last hour where someone tried to log in but couldn't."
    - "Failed authentication logs from the past hour."
3. For each query:
    - Record the CAI-generated JSON DSL.
    - Submit the DSL to Elasticsearch.
    - Extract returned event UUIDs.
    - Compare with PI-Key ground truth labeled failed-login events.
    - Compute precision, recall, and F1 score for each phrasing.
4. Stop Data Collection after all queries are issued.

Data to be Collected:

- CAI-generated JSON for each paraphrased query
- Elasticsearch returned UUID sets
- Ground-truth UUIDs from PI-Key (failed-login labels)
- Precision, recall, F1 metrics for each phrasing
- Error logs and timing data

Expected Outcomes / Success Criteria:

- All CAI-generated queries must be valid Elasticsearch JSON DSL (no schema errors, no malformed operators).
- For every paraphrased query, returned results must map to the same underlying failed-login event set.
- The average F1 across all paraphrases must be ≥ 0.95 (matching project hypothesis).

- Query structure should remain logically aligned across samples (consistent filtering of event.outcome: failure, event.category: authentication, etc.).

6.0 Exit Criteria & Teardown

The test cycle is complete when:

1. All CAI queries executed and logged.
2. All raw data copied to secured storage with hashes recorded in Lab Notebook.
3. All data collection processes terminated.
4. Kibana dashboards exported.

7.0 Roles & Responsibilities

Role	Name	Responsibility
Test Executor	Koi McFarland	Runs test cases, collects data
Reviewer	Prof. Henderson	Reviews Test Plan & results
DoD/ACE Mentor	Prof. Littleton	Provides oversight and technical guidance

8.0 References

ID #	Document Title	Location / Link
TCSP-001	Testbed Setup & Config Plan	./docs/docx/Testbed-Setup-Configuration-Plan.docx
PC-001	Project Charter	./docs/docx/Project-Charter.docx
LN-001	Lab Notebook	./docs/docx/Lab-Notebook.docx

Revision History

Ver. #	Date	Author	Summary of Change
1.1	2025-10-29	Koi McFarland	Filled out document.
1.2	2025-11-20	Koi McFarland	Redid the Test Cases
1.3	2025-12-07	Koi McFarland	Modified TC-006 natural language queries.